

Subhadeep Koley

PhD Student, Computer Science and Engineering, Lehigh University

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Research interests: aerial robotics, physical human-robot interaction, motion planning, learning-based control. Seeking research internships for Summer 2027.

EDUCATION

Lehigh University

PhD, Computer Science and Engineering. Advisor: Prof. David Saldaña

Aug 2024 – Present

Bethlehem, PA

Indian Institute of Engineering Science and Technology (IIST), Shibpur

B.Tech., Mechanical Engineering. CGPA: 8.56/10

2020 – 2024

Howrah, India

PUBLICATIONS

Under review

- **Subhadeep Koley**, Benjamin Greenberg, Yifei Simon Shao, Juan Aceros, Nadia Figueroa, David Saldaña. **HINT-Blimp: Human INTent Inference from Multimodal Cues for Robotic Blimps**. Under review, ICRA 2027. [[arXiv](#)] [[video](#)]
- **Subhadeep Koley**, Subhrajit Bhattacharya, David Saldaña. **Planning Trajectories that Bounce: Reflection Classes for Collision-Tolerant Robots**. Under review, ICRA 2027. [[arXiv](#)] [[video](#)]

Conference papers

- Yizhuo Wang, Yuhong Cao, Jimmy Chiun, **Subhadeep Koley**, Mandy Pham, Guillaume Adrien Sartoretti. **ViPER: Visibility-based Pursuit-Evasion via Reinforcement Learning**. Conference on Robot Learning (CoRL), 2024. [[paper](#)] [[video](#)] [[code](#)]
- Ajit Das, Saumendra Nath Mishra, **Subhadeep Koley**, Sourav Sarkar, Achintya Mukhopadhyay, Swarnendu Sen. **Early Detection of Thermal Runaway of Lithium-ion Battery: An Experimental Study**. IEEE Conf. on Sustainable Energy and Future Electric Transportation (SEFET), 2023. **Best Paper Award**. [[paper](#)]

Workshop papers

- **Subhadeep Koley**, Benjamin Greenberg, Yifei Simon Shao, Nadia Figueroa, David Saldaña. **HINT-Blimp: Human INTent Inference from Multimodal Cues for Robotic Blimps**. IROS 2026 Workshop on Nonverbal Cues for Human-Robot Cooperative Intelligence, Pittsburgh, 2026. Spotlight talk and poster. [[paper](#)] [[video](#)] [[poster](#)]

Posters and demos

- **HINT-Blimp**: Poster presentation, Northeast Robotics Colloquium (NERC), Princeton University, Oct 2026.
- Live demonstration of [MochiSwarm](#), an open-source swarm of lightweight robotic blimps. ICRA 2025, Atlanta.

RESEARCH EXPERIENCE

Graduate Researcher, Swarms Lab, Lehigh University

Advisor: Prof. David Saldaña

Aug 2024 – Present

Bethlehem, PA

- Working on autonomous navigation and mapping of large unknown indoor environments using Unitree Go2.
- Built a multimodal intent-inference system for robotic blimps using physical pushes, voice and pointing; combining pushes and voice identified the intended goal 86% of the time, usually within two interactions, and let users steer the blimp around obstacles only they knew about.
- Developed a planner that lets collision-tolerant robots bounce off walls deliberately, grouping paths into reflection classes by wall-contact sequence; the best bouncing path reduced control effort by a median 16.6% versus a collision-free path in simulation.
- Won first place at [Defend the Republic 2024](#) against teams including Georgia Tech, UMich and ASU; built and maintained 16+ robots and added ultrasonic wall avoidance to the hardware and software stack.

Research Intern, MARMoT Lab, National University of Singapore

Mentor: Dr. Guillaume Sartoretti

Jan 2024 – May 2024

Remote

- Worked on reinforcement learning for multi-robot visibility-based pursuit-evasion in unknown environments; published at CoRL 2024 (ViPER).

SURGE Research Intern, SPiN Lab, IIT Kanpur

May 2023 – Aug 2023

*Mentor: Dr. Ketan Rajawat**Kanpur, India*

- Part of a 10-person team building ROS software for a decentralized drone swarm, deployed on five real drones [report] [video].
- Developed a mission framework supporting leader-follower flight and independent per-drone missions [video].
- Built decentralized leader election so the swarm elects a new leader on leader failure.
- Developed inter-drone collision avoidance, tested under motion capture, and network monitoring and congestion control for the swarm's mesh network.

Research Intern, Jadavpur University

Jul 2022 – Dec 2022

*Mentor: Dr. Sourav Sarkar**Kolkata, India*

- Built a heat- and gas-sensor setup for early detection of lithium-ion battery thermal runaway, leading to a Best Paper Award at IEEE SEFET 2023.
- Built a multi-point wireless temperature acquisition system for a Tata Steel-funded project studying cooling rates in rolling-mill cooling beds.

Flight Software Intern, EndureAir Systems

Dec 2022 – Feb 2023

*Mentor: Dr. Kuldeep Kumar Dhiman**Noida, India*

- Worked with the PX4 flight stack; built a MATLAB range-extension tool for a drop tank with delayed wing deployment.

AWARDS

- [First place](#), Defend the Republic drone competition, Indiana University Nov 2024
- Best Paper Award, IEEE SEFET Aug 2023

TEACHING

- Teaching Assistant, CSE 340: Design and Analysis of Algorithms, Lehigh University

TECHNICAL SKILLS

Languages	Python, MATLAB
Robotics	ROS, PX4, Gazebo, CoppeliaSim, MoveIt, motion capture
ML / Vision	PyTorch, OpenCV
Hardware	ESP32, ATmega328, Raspberry Pi, SolidWorks, 3D printing